Roots of polynomial functions

1

a No b Yes c Yes d No

2

a

i $x=-6,\ x=3,\ x=-9$

ii $x=-6$ has the multiplicity of 3, $x=3$ has the multiplicity of 1 and $x=-9$ has the multiplicity of 1.

b

i $x=-1,\ x=6,\ x=-2$

ii $x=-1$ has the multiplicity of 2, $x=6$ has the multiplicity of 1 and $x=-2$ has the multiplicity of 5.

c

i $x=6,\ x=-2,\ x=0$

ii $x=6$ has the multiplicity of 4, $x=-2$ has the multiplicity of 5 and $x=0$ has the multiplicity of 3.

d

i $x=3,\ x=\sqrt{5},\ x=-\sqrt{5}$

ii $x=3$ has the multiplicity of 1, $x=\sqrt{5}$ has the multiplicity of 1 and $x=-\sqrt{5}$ has the multiplicity of 1.

3

a Yes b No c No d No

4

a Yes b Yes c Yes d No

e Yes f No g No h Yes

5

a $x+4,\ x+3,\ x-2,\ x-4,\ x-5$

b $h(x)=(x+4)(x+3)(x-2)(x-4)(x-5)$

c

i Yes ii No iii No iv Yes

d 0

Equations of polynomial functions

6

a $x=-1,\,-5$

b $x=-1$

c $y=(x+1)^2\,(x+5)$

7 $y=(x+4)^2\,(x-1)$

8

a $y=-4x^3+12x^2+52x-60$

b $y=-4x^3+28x^2-40x$

c $y=\dfrac{1}{4}x^3-9x$

d $y=3x^3+6x^2-57x-60$

e $y=3x^3-30x^2+87x-60$

f $y=-2x^3-4x^2+22x+24$

g $y=5x^3+40x^2+95x+60$

h $y=-5x^3+15x^2+30x-40$

9 $b=-5$

10

a $y=2\,(x+3)\,(x+1)\,(x-1)$

b $y=-\,(x+4)\,(x-2)\,(x+1)$

c $y=(x+1)\,(x-2)\,(x-4)$

d $y=-2\,(x+1)\,(x-1)\,(x-3)$

e $y=-\dfrac{1}{6}\,(x+4)\,(x+3)\,(x+2)\,(x+1)$

f $y=2\,(x+2)\,(x+1)\,(x-1)\,(x-2)$

g $y=\dfrac{1}{4}\,(x-1)\,(x-2)\,(x-3)\,(x-4)$

h $y=-2\,(x+2)\,(x+1)\,(x-1)\,(x-2)$